

■ 2.1.2 The Meaning of Expressions

The notion of expressions is a crucial unifying principle in *Mathematica*. It is the fact that every object in *Mathematica* has the same underlying structure that makes it possible for *Mathematica* to cover so many areas with a comparatively small number of basic operations.

Although all expressions have the same basic structure, there are many different ways that expressions can be used. Here are a few of the interpretations you can give to the parts of an expression.

Meaning of <i>f</i>	Meaning of <i>x</i> , <i>y</i> , ...	Examples
Function	arguments or parameters	<code>Sin[x]</code> , <code>f[x, y]</code>
Command	arguments or parameters	<code>Expand[(x + 1)^2]</code>
Operator	operands	<code>x + y</code> , <code>a = b</code>
Head	elements	<code>{a, b, c}</code>
Object type	contents	<code>RGBColor[r, g, b]</code>

Some interpretations of parts of expressions.

Expressions in *Mathematica* are often used to specify operations. So, for example, typing in `2 + 3` causes 2 and 3 to be added together, while `Factor[x^6 - 1]` performs factorization.

Perhaps an even more important use of expressions in *Mathematica*, however, is to maintain a structure, which can then be acted on by other functions. An expression like `{a, b, c}` does not specify an operation. It merely maintains a list structure, which contains a collection of three elements. Other functions, such as `Reverse` or `Dot`, can act on this structure.

The full form of the expression `{a, b, c}` is `List[a, b, c]`. The head `List` performs no operations. Instead, its purpose is to serve as a “tag” to specify the “type” of the structure.

You can use expressions in *Mathematica* to create your own structures. For example, you might want to represent points in three-dimensional space, specified by three coordinates. You could give each point as `point[x, y, z]`. The “function” `point` again performs no operation. It serves merely to collect the three coordinates together, and to label the resulting object as a `point`.

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; <http://www.wolfram.com/>; 1-800-441-6284.

© 1988 Wolfram Research, Inc. Permission is hereby granted for web users to make one paper copy of this page for their personal use. Further reproduction, or any copying of machine-readable files (including this one) to any server computer, is strictly prohibited.

You can think of expressions like `point[x, y, z]` as being “packets of data”, tagged with a particular head. Even though all expressions have the same basic structure, you can distinguish different “types” of expressions by giving them different heads. You can then set up transformation rules and programs which treat different types of expressions in different ways.